AI - Agents & Environments

AI

Swipe

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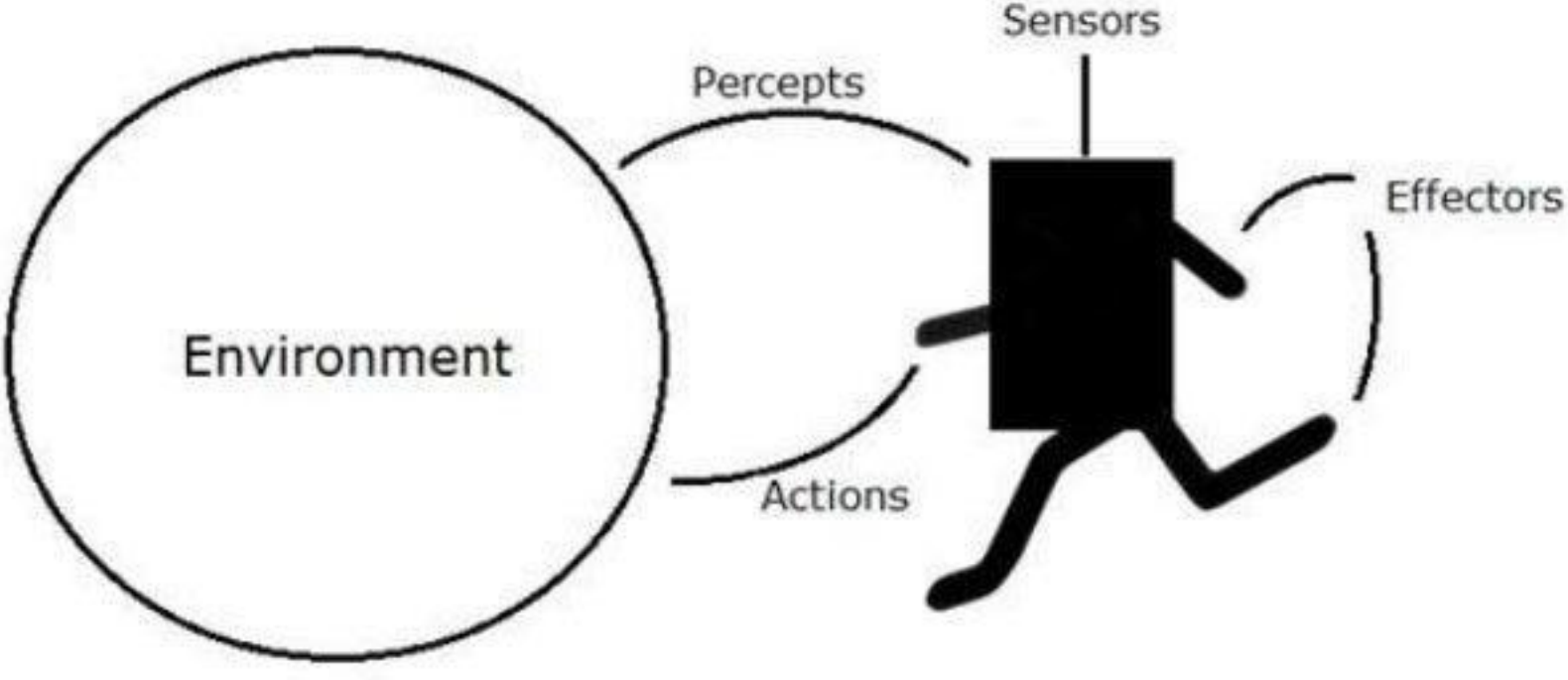
- An agent is anything that can perceive its environment through sensors and acts upon that environment through effectors.
    - A human agent has sensory organs such as eyes, ears, nose, tongue and skin parallel to the sensors, and other organs such as hands, legs, mouth, for effectors.
    - A robotic agent replaces cameras and infrared range finders for the sensors, and various motors and actuators for effectors.
    - A software agent has encoded bit strings as its programs and actions.

- **An AI system is composed of an agent and its environment. The agents act in their environment. The environment may contain other agents.**



Agent Terminology

- Performance Measure of Agent − It is the criteria, which determines how successful an agent is.
- Behavior of Agent − It is the action that agent performs after any given sequence of percepts.
- Percept − It is agent's perceptual inputs at a given instance.
- Percept Sequence − It is the history of all that an agent has perceived till date.
- Agent Function − It is a map from the precept sequence to an action.

Rationality

- Rationality is nothing but status of being reasonable, sensible, and having good sense of judgment.
- Rationality is concerned with expected actions and results depending upon what the agent has perceived.
- Performing actions with the aim of obtaining useful information is an important part of rationality.

What is Ideal Rational Agent?

- An ideal rational agent is the one, which is capable of doing expected actions to maximize its performance measure, on the basis of:-
  - Its percept sequence
  - Its built-in knowledge base
- Rationality of an agent depends on the following:-
  - The performance measures, which determine the degree of success.
  - Agent's Percept Sequence till now.
  - The agent's prior knowledge about the environment.
  - The actions that the agent can carry out.

The Structure of Intelligent Agents

- Agent's structure can be viewed as:-
  - Agent = Architecture + Agent Program
  - Architecture = the machinery that an agent executes on.
  - Agent Program = an implementation of an agent function.

- Simple Reflex Agents:-
  - They choose actions only based on the current percept.
  - They are rational only if a correct decision is made only on the basis of current precept.
  - Their environment is completely observable.

Properties of Environment

- The environment has multifold properties:-
  - Discrete / Continuous
  - Observable / Partially Observable
  - Static / Dynamic
  - Single agent / Multiple agents
  - Accessible / Inaccessible
  - Deterministic / Non-deterministic
  - Episodic / Non-episodic

Topics for next Post

- AI - Popular Search Algorithms
- AI - Natural Language Processing
- Artificial Intelligence - Expert Systems

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